EC 1745 Midterm Notes v10 Name: _____

EXCEL BASICS

- Every formula starts with **=**. Math: + – * / and ^ (power). Example: **=B2-B3**, **=B5*(1+B6)^2**.
- Point at cells instead of retyping numbers. **$** locks a cell when you copy a formula (**=B2/$B$20**); press F4 to add it.
- Rates as decimals: 5% = 0.05; 100 bp = 0.01. Dates: **=DATE(2025,10,7)**. Copy right/down: Ctrl+R / Ctrl+D.
- Type **=PV(** and Excel shows the inputs in order. Show as %: Home ▸ % button.

INCOME STATEMENT (top line → bottom line)

Labels in column A, one year per column (B = 2024, C = 2023). "type" = number given in the question.

#	Column A (label)	Column B
1	Year →	2024
2	**Revenue (sales)** ← TOP LINE	type
3	– Cost of goods sold (COGS)	type
4	**= Gross profit**	**=B2-B3**
5	– SG&A (operating costs)	type
6	**= EBITDA**	**=B4-B5**
7	– Depreciation & amortization	type
8	**= EBIT** (operating income)	**=B6-B7**
9	– Interest expense	type
10	**= Pretax income** (EBT)	**=B8-B9**
11	– Taxes (tax rate × EBT)	**=0.21*B10**
12	**= Net income** ← BOTTOM LINE	**=B10-B11**
13	– Dividends	type
14	**= Retained earnings** (kept)	**=B12-B13**

EPS = NI / shares. Retention ratio **b** = (NI – dividends) / NI.

BALANCE SHEET (snapshot on one date)

ASSETS	LIABILITIES & EQUITY
Cash	Accounts payable (AP)
+ Accounts receivable (AR)	+ Short-term debt
+ Inventory	**= Current liabilities**
= Current assets	+ Long-term debt
+ Gross PP&E	**= Total liabilities**
– Accum. depreciation	+ Shareholders' equity
= Net PP&E	
+ Other assets	
= TOTAL ASSETS	**= TOTAL LIAB. + EQUITY**

- **Total assets = total liabilities + equity**, always. One number missing (often cash)? Cash = total L+E – all other assets.
- Equity this year = equity last year + retained earnings + new stock – buybacks.
- Debt = short-term + long-term debt (AP is not debt). Only debt and equity given? Total assets = debt + equity.
- New cash balance: subtract AR, inventory and net PP&E from TL + equity.
- **NWC = current assets – current liabilities.** Notes payable = short-term debt (current liability). Patents, copyrights = intangible fixed assets. Equity = common stock + accumulated retained earnings.

CASH FLOW STATEMENT (cash ≠ net income)

Δ (delta) = change = this year – last year. Example: ΔAR = AR 2025 – AR 2024.

1. ΔNWC	ΔNWC = ΔAR + ΔI – ΔAP (ΔI = Δinventory)
2. CFO (operations)	CFO = Net income + Depreciation – ΔNWC
3. CapEx	CapEx = ΔNet PP&E + depreciation
4. CFI (investing)	CFI = – CapEx
5. CFF (financing)	CFF = ΔDebt + new stock – dividends
6. Check (vs. BS)	**CFO + CFI + CFF = change in cash** ✓

Signs: asset ↑ uses cash (subtract); liability ↑ gives cash (add). Depreciation is added back (no cash leaves).
Ex: AR +30, inventory +30, AP +20 ⇒ ΔNWC = 30 + 30 – 20 = **40** ⇒ CFO = NI + Dep – 40

Free cash flow = EBIT × (1 – t) + Dep – CapEx – ΔNWC, where ΔNWC = ΔAR + ΔInv – ΔAP

CASH FLOW FROM ASSETS (textbook, Ch. 2)

1. OCF	OCF = EBIT + Depreciation – Taxes
2. NCS (net cap. spending)	NCS = ending net PP&E – beginning + Dep
3. ΔNWC (cash included)	ΔNWC = (CA – CL)end – (CA – CL)begin
4. CF from assets	CFFA = OCF – NCS – ΔNWC
5. CF to creditors	Interest – net new LT debt
6. CF to stockholders	Dividends – net new equity
7. Check	**CFFA = creditors + stockholders** ✓

Debt redeemed ⇒ net new borrowing is negative: CF to creditors = interest + amount redeemed. New stock issued ⇒ subtract from dividends.
Backward: ΔNWC = OCF – net capital spending – (CF to creditors + CF to stockholders).

RATIOS (use year-end numbers)

Gross margin	Gross profit / Revenue
Operating margin	EBIT / Revenue
Profit margin	Net income / Revenue
ROE	Net income / Equity
ROA	Net income / Total assets
Asset turnover	Revenue / Total assets
Equity multiplier	Total assets / Equity
EM shortcut	= 1 + Debt/Equity
DuPont	**ROE = Profit margin × Asset turnover × Equity multiplier**
Sustainable growth	ROE × b (or ROE·b / (1 – ROE·b)) Textbook uses ROE·b / (1 – ROE·b) b = retention ratio = retained earnings / net income, or 1 – dividend payout ratio
Internal growth	ROA × b (or ROA·b / (1 – ROA·b)) Textbook uses ROA·b / (1 – ROA·b)
Current ratio	Current assets / Current liabilities
LT debt ratio	LT debt / (LT debt + Equity)
Receivables turnover	Sales / Receivables
Days' sales in receivables	Receivables / (Sales / 365) = 365 / receivables turnover
Interest coverage (TIE)	EBIT / Interest expense
Debt-to-equity	Total debt / Equity (textbook: all liabilities)

Profit margin = profitability; turnover = efficiency; multiplier = leverage (raises ROE and risk).
Margin ≠ return. Profit margin = profit per $ of sales. ROA / ROE = profit per $ invested. A low margin can still earn a good return if turnover is high.
DuPont backward: solve ROE = PM × TAT × EM for the missing piece (NI = ROE × Equity; Equity = Assets / EM; Sales = Receivables × 365 / days).

WHAT THE FIRM IS WORTH

Market cap = shares × share price (= market value of equity)
Enterprise value (EV) = market cap + debt (ST + LT) – excess cash
Given EV (e.g. EV = 8 × EBIT): equity value = EV – debt + cash
P/E = market cap / net income (= price / EPS). EV/EBITDA = EV / EBITDA

HOW TO ANSWER A RATIO QUESTION

1. Type the statements into Excel (templates above).
2. Put each ratio in its own labeled cell, as a formula pointing at the statement cells.
3. Format as % where it is a percent. Check that the DuPont product equals ROE.
4. Asked to interpret? Say which number rose or fell, and why it matters.

TIME VALUE OF MONEY (draw a timeline; t = 0 is today)

1. One cash flow	PV = CF / (1+r)^t FV = PV × (1+r)^t
2. Annuity (C, t = 1…N)	PV = C/r × (1 – 1/(1+r)^N)
3. Growing annuity	PV = C_1/(r – g) × (1 – ((1+g)/(1+r))^N)
…in Excel	**=PV((1+r)/(1+g)-1, N, -C_1/(1+g))**
4. Perpetuity	PV = C/r
5. Growing perpetuity	PV = C_1/(r – g)
6. Annuity due	PV × (1+r) if 1st payment is today
7. Value at year T	Value at T = PV × (1+r)^T

Compare offers at the **same date**; bigger PV wins.

By hand ("dumb"): rate in B1; row 3 = years 0…N; row 4 = cash flows.

Row 5: discount factor	**=1/(1+$B$1)^B3** (copy right)
Row 6: PV of each CF	**=B4*B5** (copy right)
Total PV	**=SUM(row 6)**

=PV(r,n,-C)	Today's value of n payments of C
=FV(r,n,-C)	Value at year n
=PMT(r,n,-PV)	Payment that pays off PV
=RATE(n,C,-PV)	Discount rate implied by the deal
=NPER(r,C,-PV)	Number of payments
=NPV(r,C1:Cn)	Starts at t = 1! Add CF0 yourself
=IRR(C0:Cn)	Rate where NPV = 0 (C0 negative)
=RATE(n,0,-PV,FV)	Rate for one lump sum (no payments)
=NPER(r,0,-PV,FV)	Periods for one lump sum

Lump sum by hand: r = (FV/PV)^(1/n) – 1; n = LN(FV/PV) / LN(1+r). Money arrives later? Add the waiting time. Match the period: "per 3 months" over 24 months = 8 periods.

Minus signs: Excel needs money paid and money received to have opposite signs. Payments start today? Add **,0,1** at the end: **=PV(r,n,-C,0,1)**.

INTEREST RATES & LOANS

APR = quoted annual rate. Per-period rate = APR / periods per year.

EAR = (1 + APR/k)^k – 1 = **=EFFECT(APR,k)**. Bond EAY = (1 + y/2)^2 – 1.

Monthly loan payment: **=PMT(APR/12,years*12,-loan)**. Balance after k payments: **=PV(APR/12,N-k,-payment)**.

Restructuring: the new deal must have the same PV at the same rate. Solve for the unknown with PMT / RATE / NPER.

BOND WORDS

Par (face) = 100, repaid at maturity. **Coupon rate** = % of par paid per year, in 2 halves (5% → 2.50 every 6 months).

Yield (YTM) = discount rate that makes PV of the bond's cash = price. Quoted as APR; use y/2 per half-year.

Yield ↑ ⇒ price ↓. Coupon > yield ⇒ price > 100. Issued at par ⇒ coupon = yield.

Corporate yield = Treasury yield + spread. 1 bp = 0.01% = 0.0001.

Basis: 0 = 30/360 (corporates), 1 = actual/actual (Treasuries). Settlement = date you buy.

BONDS WITHOUT DATES ("dumb")

No dates given? Use RATE & PV.

Price	**=PV(yield/2,2*years,-coupon*100/2,-100)**
Yield (YTM)	**=2*RATE(2*years,coupon*100/2,-price,100)**
EAY	**=(1+yield/2)^2-1**
Coupon rate	**=2*PMT(yield/2,2*years,-price,100)/100**

Price needs the yield. YTM needs the price (per 100, e.g. 87.91). EAY needs only the YTM. years = years LEFT.

Bond exchange: the new bond must be worth what the old one is worth. Solve for the new coupon with PMT. Face needed to raise $X = X / (price/100).

Annual coupons? Use the full yield and the number of years (no /2, no ×2) and the full annual coupon. **Par of 1,000 (or ¥100,000)?** Put the par where the formulas have 100; coupon = rate × par. "104.352% of par" = 1.04352 × par.

Market value of all bonds = face outstanding × price / 100. Example: 400 × 87.91 / 100 = 351.62.

BONDS WITH DATES ("smart"), STEP BY STEP

Dates given? Use YIELD & PRICE, not RATE.

1. Type inputs: settle **=DATE(…)**, maturity **=DATE(…)**, coupon, price or yield, 2, basis.
2. Treasury yield: **=YIELD(settle,mat,coupon,price,100,2,1)** (the YTM)
3. Corporate yield = Treasury yield + spread.
4. Clean price: **=PRICE(settle,mat,coupon,yield,100,2,0)**
5. Days since last coupon **=COUPDAYBS(settle,mat,2,0)**; days in period **=COUPDAYS(settle,mat,2,0)** (180 on 30/360).
6. **Accrued interest** = coupon×100/2 × COUPDAYBS / COUPDAYS
7. **Dirty price = clean + accrued.** Dealer QUOTES clean; you PAY dirty.
8. **=DURATION(settle,mat,coupon,yield,2,basis)** and **=MDURATION(…same…)**
9. % price change ≈ –MDURATION × Δyield (+100 bp = +0.01).
10. Actual: new dirty = PRICE at the new yield + same accrued, vs. old dirty price. **% change = new / old – 1.**

Also: **COUPPCD** / **COUPNCD** = previous/next coupon date (format as date); **COUPNUM** = coupons left.

DURATION (what, why, how)

What: the average time (in years) until you get the bond's cash, weighted by PV.

Why: it tells how much the price moves when yields move (interest-rate risk). Higher duration = more risk. Used to compare bonds and to match assets to liabilities.

1. Macaulay D	D = Σ (t × PV of CF) / price (t in years)
2. Modified MD	MD = D / (1 + y/2)
3. Estimated % ΔP	≈ –MD × Δy (MD 7.7, yield +1% ⇒ ≈ –7.7%)
4. Actual % ΔP	new price / old price – 1
5. From two prices	D ≈ –(% change in price) / (change in yield)

No dates given? **=DURATION(DATE(2000,1,1),DATE(2000+years,1,1),c,y,2)**

By hand: one row per coupon k = 1…N. Row 2 formulas, copied down:

A: k (period)	1, 2, … N
B: years	**=A2/2**
C: cash	**=c*100/2** (+100 in last row)
D: PV	**=C2/(1+y/2)^A2** (y = your yield cell)
e.g.	**=C2/(1+$B$8/2)^A2** ($B$8 = yield)
E: years × PV	**=B2*D2**

Price = SUM(D). **D = SUM(E) / SUM(D).** Between coupon dates, periods = k – 1 + COUPDAYSNC/COUPDAYS.

Zero-coupon D = maturity. Longer maturity ⇒ higher D. Higher coupon ⇒ lower D.

Convexity: the real price change beats the estimate (falls less, rises more).

WHICH TOOL WHEN (what you have → what to use)

Lump sum → PV / FV	CF / (1+r)^t; CF × (1+r)^t
Lump sums → rate / time	**=RATE(n,0,-PV,FV)**; **=NPER(r,0,-PV,FV)**
Level payments → PV / FV	**=PV(r,n,-C)**; **=FV(r,n,-C)**
Loan → payment; → rate	**=PMT(r,n,-PV)**; **=RATE(n,C,-PV)**
Uneven CFs → PV; → rate	**=CF0+NPV(r,CF1:CFn)**; **=IRR(…)**
Growing / forever	Growing annuity; C/r; C_1/(r – g)
First payment today	Add **,0,1** or PV × (1+r)
APR → EAR	**=EFFECT(APR,k)** = (1+APR/k)^k – 1
No dates: yield → price	**=PV(y/2,2*yrs,-c*100/2,-100)**
No dates: P → YTM / cpn	**=2*RATE(…)**; **=2*PMT(…)/100**
Dates: price → yield	**=YIELD(s,m,c,P,100,2,b)**
Dates: yield → price	**=PRICE(…)** clean; + accrued = dirty
Treasury + spread	YIELD(Treasury) + spread → PRICE
Duration	**=DURATION(…)** / **=MDURATION(…)**
Two prices → duration	–(% change in price) / Δy
Yield moves → % ΔP	Est. –MD × Δy; actual new/old – 1
Raise $X; all bonds	Face = X/(P/100); MV = face × P/100
Restructure / exchange	Same PV; solve with PMT / RATE / NPER
Balance sheet item missing	Plug with A = L + E
EV multiple given	Eq = EV – debt + cash; P/E = Eq / NI
Why did ROE change?	DuPont: PM × TAT × EM